

Chapter 3: What is Problem Solving?

Introduction

Problem Solving is the process of understanding a problem, analyzing the given information, finding a logical solution, and converting that solution into code.

Definition

Problem Solving means:

Input → Problem → Output

Example:

Input: 5 7

Output: 12

Your brain thinks: "How can I convert the input into the output?" This thinking process is called Problem Solving.

The Complete Problem-Solving Process

Step 1: Read Carefully

Example: Find the largest number among 3 numbers.

Input: 10 5 8

Output: 10

Read the question carefully and understand what is being asked.

Step 2: Identify Input

Ask: What information is provided?

Here, three numbers are given: 10, 5, and 8.

Step 3: Identify Output

Ask: What should be returned?

The answer should be the largest number.

Step 4: Find Logic

Think like a detective.

Compare:

10 > 5

10 > 8

Therefore, 10 is the largest.

Step 5: Write Steps

1. Take three numbers.
2. Compare them.
3. Find the largest.
4. Print the answer.

These steps become the algorithm.

Step 6: Convert to Code

Once the logic is clear, write the solution in a programming language.

How Good Programmers Think

Beginners think:
What code should I write?

Experts think:
What logic should I use?

Code is simply the translation of logic.

Why Problem Solving Matters

- Improves logical thinking
- Makes coding easier
- Helps in interviews
- Builds confidence
- Forms the foundation of DSA

Problem Solving Formula

Understand Problem → Identify Input → Identify Output → Find Logic → Create Algorithm → Write Code

Key Takeaways

- Every program solves a problem.
- Logic is more important than syntax.
- Algorithms are step-by-step solutions.
- Strong problem-solving skills lead to strong DSA skills.

Remember:

"A good programmer is not the one who knows the most syntax, but the one who can solve the most problems."